

Who to focus on, why, and with what conversation

KavionLens360 turns 1,048,567 bank transactions into one analytical view per customer, groups 841,543 customers into five behaviour segments, and ranks them by transparent cross-sell *signals* so a relationship manager knows whom to call first.

Public Kaggle data from an Indian bank (2016), not ICICI Bank data. Analysis window 01 Aug – 15 Sep 2016, as of 16 Sep 2016.
Every number on this page is generated from the pipeline.

Customers

8.42 L

841,543 CustomerIDs

Transactions analysed

9.89 L

₹155 Cr moved

Value from top 10%

63%

Gini 0.74

Behaviour segments

5

silhouette 0.28 · ARI $\geq$ 0.93

Tier A priority customers

42.1K

₹1,721 Cr addressable
balance

1. The problem

Banks store activity as transactions, but managers think in customers. With limited relationship-manager time, a raw transaction log cannot say **who to focus on, why, with what offer, where they are, and how large the opportunity is**.

This project builds a small analytical version of a Customer 360: a reconciled customer table, behaviour segments with a named action each, and a ranked opportunity list, delivered through SQL, Python, Power BI, an Excel MIS and an executive deck.

Value chain: data → analysis → insight → business action.

- 1 PostgreSQL 16: staging, 17 cleaning rules, star model, `customer_360`
- 2 Python: profiling, EDA, statistics, K-Means segmentation, opportunity scoring
- 3 Power BI (5 pages), Excel MIS (7 sheets), deck (8 slides), this site
- 4 Reconciliation: SQL = Python = Excel = Power BI, variance 0

2. Data, profiling and quality

Profiling the 1,048,567-row file confirmed every published count, and found two issues the dataset page does not mention:

Coverage ends on 15 September

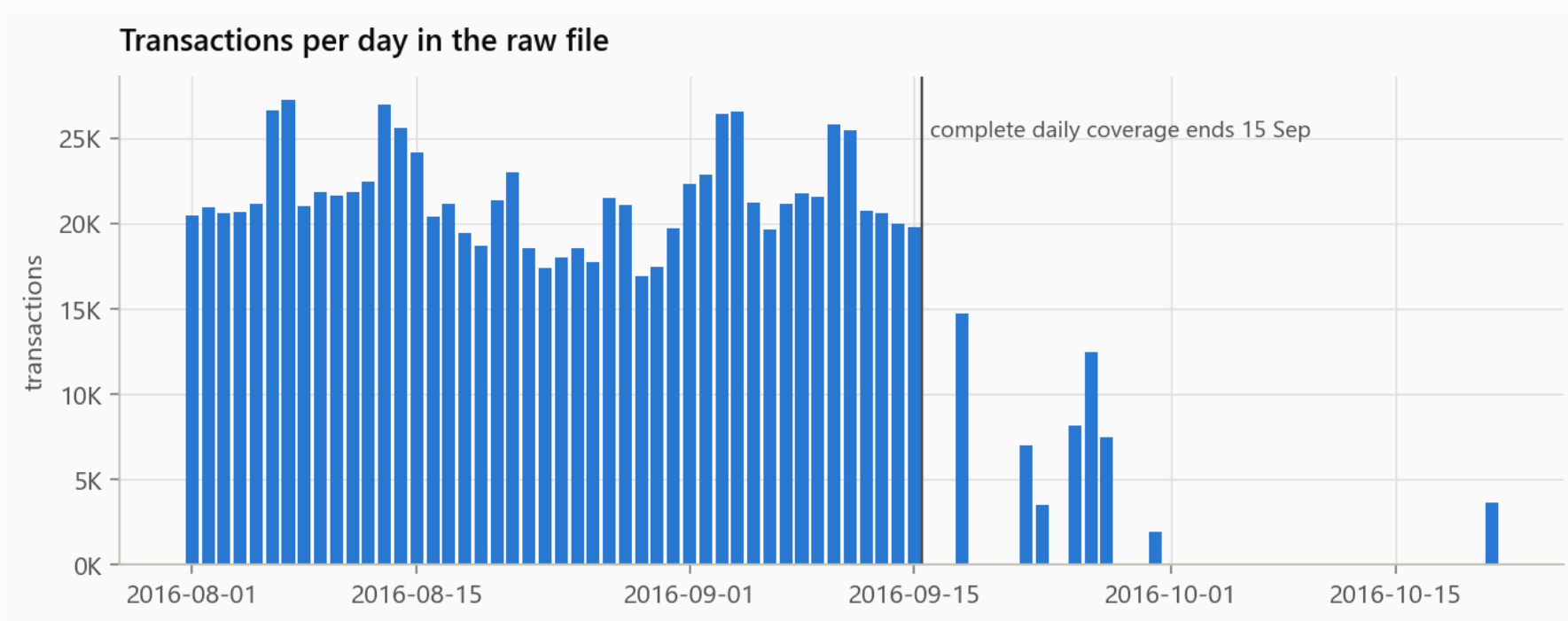
Every day from 1 Aug to 15 Sep has roughly 17–27K transactions; after that the file holds only a few scattered days. Because recency and “active” rules need complete coverage, the analysis window was fixed to 46 days and the sparse tail (58,826 rows) was logged, not silently dropped.

CustomerID is not a stable person key

Only 0.9% of repeat CustomerIDs keep the same birth date, gender and city across their rows. The ID is kept as the key (as specified), conflicts are resolved by the most-frequent-then-latest rule, and repeat-customer features are interpreted with care.

Row waterfall	Rows
Raw rows in staging	1,048,567
Rejected: PARTIAL_COVERAGE_PERIOD	-58,826
Rejected: ZERO_AMOUNT	-794
Clean rows	988,947

Quality scorecard: ✓ 17 pass ! 1 documented exception i 6 findings . The one exception is usable birth dates (94%), because of the `1/1/1800` placeholder. 9,355 raw location spellings were standardised into 8,773 cities.



Transactions per day in the raw file. Complete daily coverage ends on 15 Sep 2016.

3. Customer 360 insights

Value is extremely concentrated

The top 1% of customers generate 27% of transaction value, the top 10% generate 63% and hold 77% of balances (Gini 0.74).

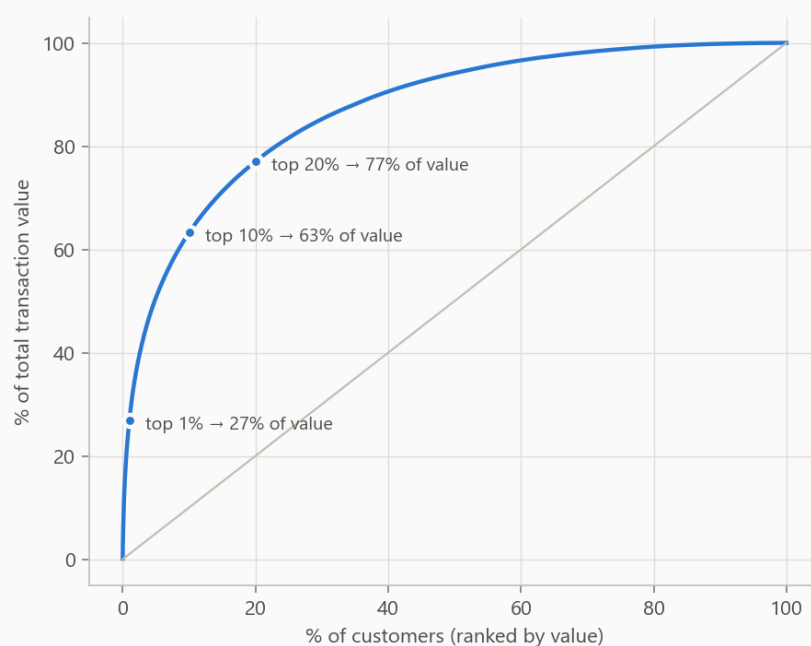
Metros carry the book

Mumbai, New Delhi, Bangalore, Gurgaon, Delhi hold 42% of value. Weekend days run 22% above weekdays, and 63% of activity falls between 12:00 and 21:00.

Most customers appear once

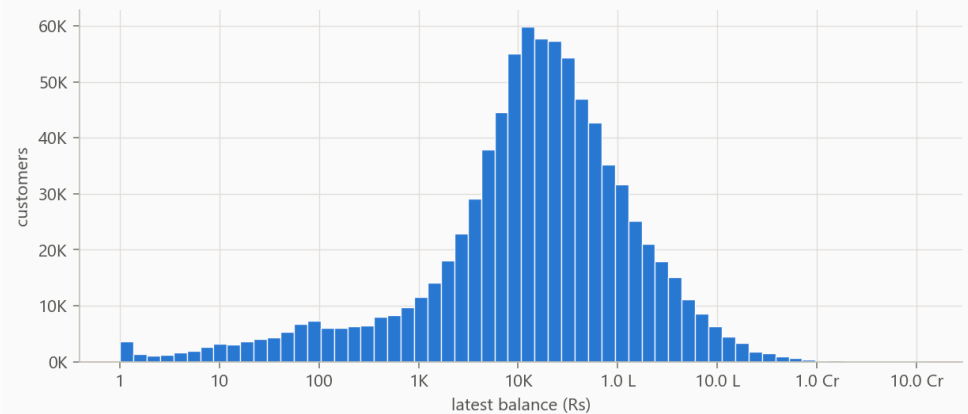
84.6% of customers made a single transaction in the window, so frequency is weak evidence for most of the base. Every list carries an evidence level.

How concentrated is value? Cumulative share of transaction value

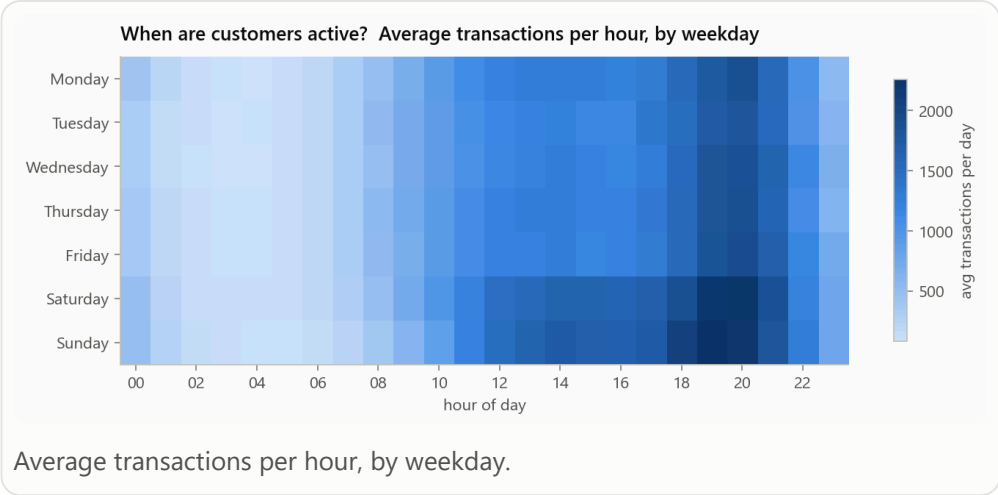
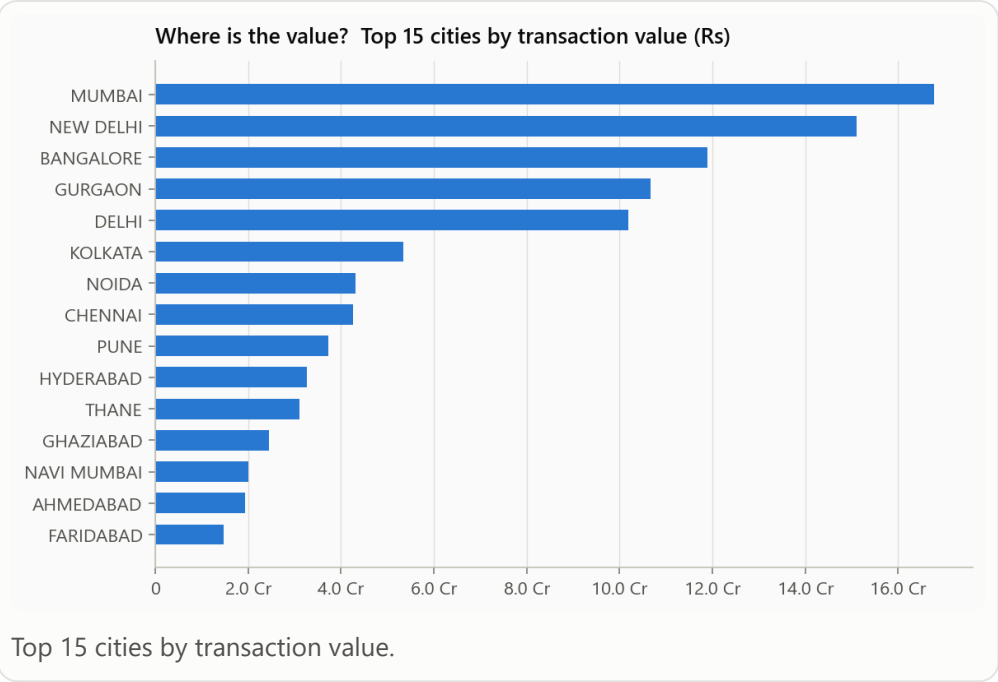


Cumulative share of transaction value by customers ranked by value.

How is wealth spread? Latest account balance per customer (log scale)



Latest balance per customer, log scale.



Statistics, with effect sizes

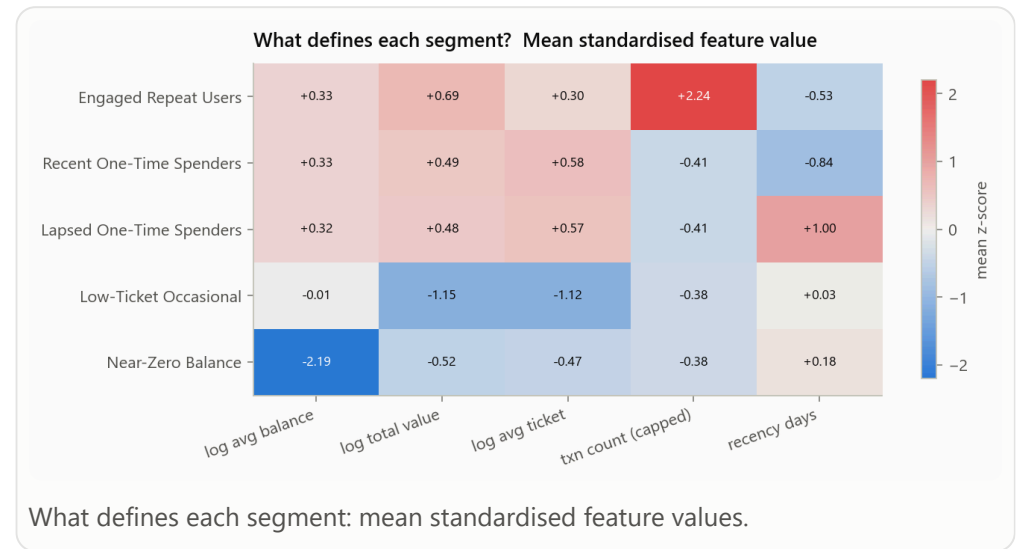
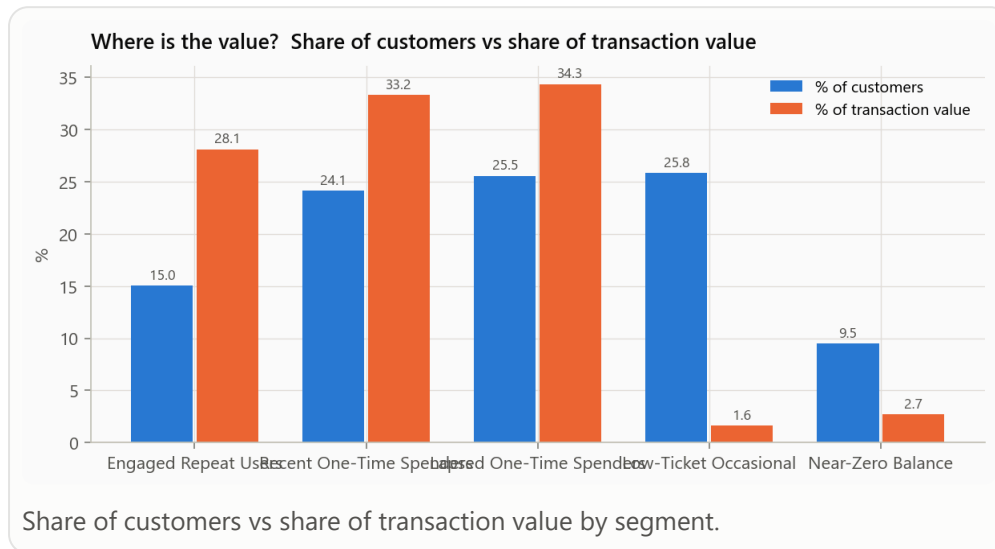
With 840K customers every test is “significant”, so each reports an effect size. The result is consistent: **behaviour, not demographics, explains value**. Gender barely moves ticket size (rank-biserial $r = 0.08$), age explains little of balance ($\epsilon^2 = 0.06$), metro status barely changes the high-value share (Cramér's $V = 0.05$), while balance and transaction value are only moderately linked (Spearman $\rho = 0.30$).

Test	Question	Effect size	Detail
Mann-Whitney U	Avg ticket differs between men and women	rank-biserial $r = 0.079$	median M Rs 473 vs F Rs 560
Kruskal-Wallis	Avg balance differs across age groups	$\epsilon^2 = 0.062$	medians: 18-24 Rs 9,741, 25-34 Rs 17,614, 35-44 Rs 31,889, 45-54 Rs 44,941, 55-64 Rs 52,414, 65+ Rs 54,368
Spearman	Balance is associated with transaction value	$\rho = 0.296$	
Chi-square	High-value share differs by city tier	Cramer's $V = 0.052$	high-value %: Metro satellite 18.8%, Other 15.5%, Tier 1 metro 19.8%

4. Five behaviour segments

K-Means on five behaviour features (log balance, log value, log ticket, capped frequency, recency), with k chosen from 3–10 by elbow, silhouette, Calinski-Harabasz and five-seed stability. k = 5: silhouette 0.279, ARI ≥ 0.93 , smallest segment 9.5%. Gender and city were excluded from the inputs and only used to profile.

Segment	Customers	Value	Median balance	Txns / customer	Active	Action
Engaged Repeat Users	15.0%	28.1%	₹29,937	2.14	59%	Deepen the relationship: RM introduction call, needs review, bundle offer
Recent One-Time Spenders	24.1%	33.2%	₹29,017	1.00	73%	Convert to repeat use: welcome-back nudge and digital activation while the visit is recent
Lapsed One-Time Spenders	25.5%	34.3%	₹28,730	1.00	0%	Win-back: service call or message to understand needs; offer relevant product review
Low-Ticket Occasional	25.8%	1.6%	₹12,443	1.01	33%	Low-cost digital nurture only; no RM time
Near-Zero Balance	9.5%	2.7%	₹86	1.01	29%	Service check and activation; exclude from cross-sell lists



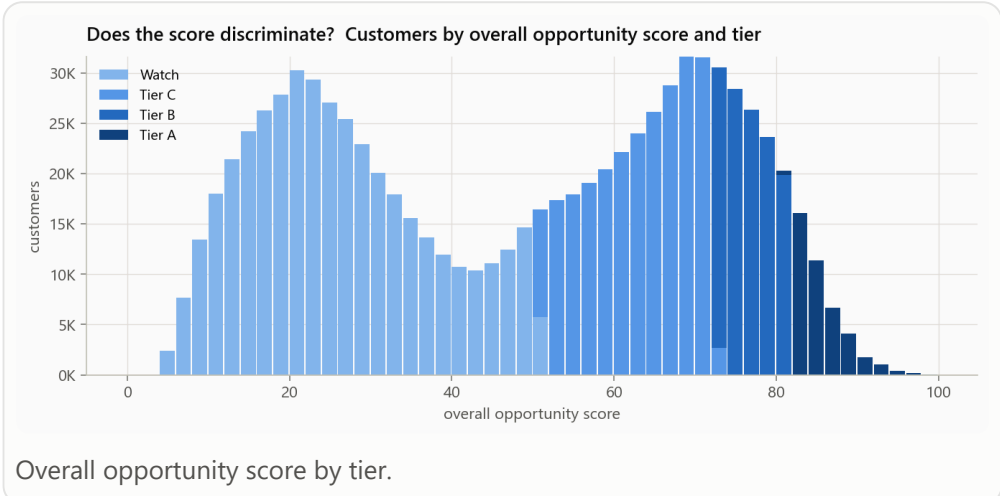
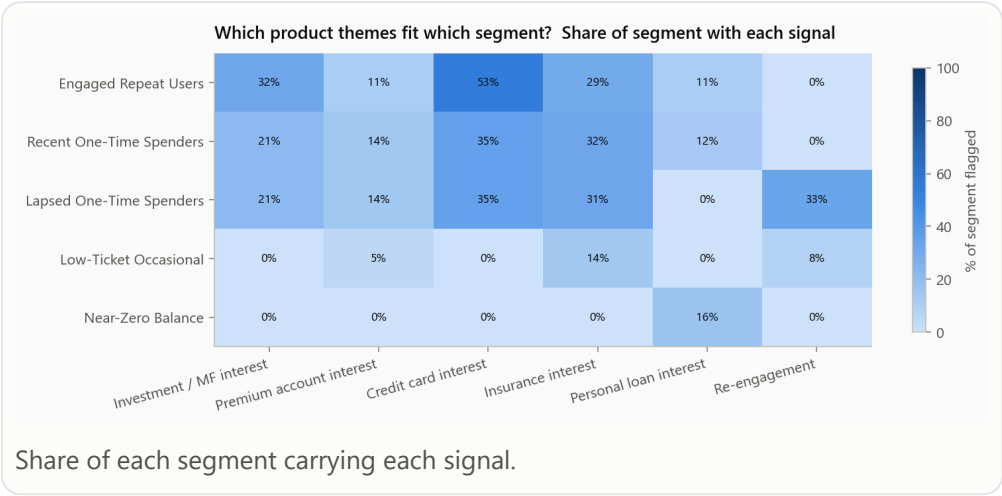
Reading: Engaged Repeat Users are 15% of customers but 28% of value. The two one-time spender segments look alike in value and differ only in recency; together they hold 68% of value. The two low-value segments (35% of customers) contribute 4.4% of value and belong in low-cost digital channels.

5. Cross-sell opportunity signals

Signals inferred from behaviour; product ownership is unknown. Six transparent rules use percentile thresholds. Each customer gets a score: $100 \times (0.35 \text{ value} + 0.25 \text{ engagement} + 0.30 \text{ rule fit} + 0.10 \text{ recency})$, and tiers A (top 5%), B (next 15%), C (next 30%) and Watch. These rank conversations; they are not eligibility or credit decisions.

Tier	Customers	Share	Min score	Addressable balance
A	42,077	5%	82.0	₹1,721 Cr
B	126,231	15%	72.2	₹3,718 Cr
C	252,463	30%	50.7	₹3,513 Cr
Watch	420,772	50%	4.4	₹42.8 Cr

Signal	Customers	% of base
Investment / MF	126,981	15.1%
Premium account	83,991	10.0%
Credit card	215,414	25.6%
Insurance	199,478	23.7%
Personal loan (demand only)	51,725	6.1%
Re-engagement (service)	87,724	10.4%



Robustness: moving any weight by ± 10 points keeps 84–100% of the top 1,000 customers (rank correlation ≥ 0.99), so the judgement-based weights do not drive the list. **Caveat:** 48% of customers carry at least one signal, which is broad; the credit-card and insurance rules are the first to tighten once response data exists.

6. From insight to action

Output	What we found	Why it matters	What a manager should do
Value concentration	Top 10% $\rightarrow$ 63% of value; top 1% $\rightarrow$ 27%	Effort pays off unevenly	Named RM coverage for the top decile
Segment profiles	Engaged Repeat Users: 15% of customers, 28% of value	Different needs per segment	Deepen this group first: needs review, cards and investments
Lapsed one-time spenders	26% of customers, 34% of value, not seen for 2+ weeks	Valuable but going quiet	Re-engagement call before any sale
Low-value tail	35% of customers, 4.4% of value	RM time has little return	Digital nurture only; activation checks for near-zero balances
Opportunity tiers	Tier A: 42,077 customers, ₹1,721 Cr	Limited RM time	Work Tier A first, review weekly, measure response
Geography	Top 5 cities hold 42% of value	Effort allocation	Staff metro hubs; use the city scorecard for high value-per-customer smaller cities

Open the Manager Action view $\rightarrow$ who, why, what, where and how large, on one screen.

7. Limitations and responsible use

- Public data from one bank in 2016 and a 46-day window: no seasonality, trend or tenure claims.
- No product holdings, revenue or campaign outcomes: opportunity is a signal, value is a proxy (balance and activity).
- CustomerID is not a stable person key in this extract; repeat behaviour is weak evidence.
- Clusters describe behaviour; they do not predict it.
- Age is used only as a life-stage hint; gender and city are never model inputs. Flags start conversations; they never decide eligibility.

Version 2

- Add product holdings and campaign response to build propensity models and measure lift.
- Longer time window and entity resolution for a reliable customer key.
- Segment-drift monitoring and fairness checks on every refresh.

8. Deliverables

PDF Analytical report This page as a printable PDF	PDF Executive deck 8-slide management presentation	PPTX Executive deck (editable) PowerPoint with native charts	XLSX Excel MIS 7 sheets: XLOOKUP, SUMIFS, PivotTables, Power Query
PDF Excel MIS (print) The MIS workbook as PDF	PDF Power BI report 5-page Power BI export (coming soon)		

Code

SQL Pipeline SQL Staging, cleaning, model, Customer 360, DQ	SQL 26 analytical questions CTEs and window functions	IPYNB Notebooks Profiling, EDA and statistics, segmentation, scoring	PBIP Power BI project TMDL model + PBIR report, text-based
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KavionLens360 · Bank Customer 360 & Segmentation Analytics · portfolio project by Y. Rudraaksh Reddy · built 03 Oct 2026 from github.com/rudraakshreddy/kavionlens360.
Data: public Kaggle dataset “Bank Customer Segmentation (1M+ Transactions)”, an Indian bank, 2016. It is not ICICI Bank data and nothing here describes any bank's internal systems. Opportunity flags are behavioural signals, not product recommendations or credit decisions.